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Date: 29/OCT/2026

Author: Brett Hallen

Step	Command	Code	Note
001	f CLEAR	137	Clear all registers (last burn = 0)
002	5	005	Starting height
003	0	000	
004	0	000	
005	STn	300	Store height in R0
006	0	000	
007	5	005	Starting velocity (down = +)
008	0	000	
009	STn	300	Store velocity in R1
010	1	001	
011	1	001	Starting fuel
012	2	002	
013	0	000	
014	STn	300	Store fuel in R2
015	2	002	
016	5	005	Lunar gravity
017	STn	300	Store gravity in R9
018	9	011	
019	STn	300	Safe landing speed (5) in R8
020	8	010	

Register	Purpose	Register	Purpose
0	Lander height	6	
1	Velocity (down = +)	7	
2	Fuel remaining	8	Safe landing speed
3	Burn (also last burn)	9	Lunar gravity
4	Net acceleration A = G - B	.	
5	Scratch (velocity squared)	CHG SIGN	

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Step	Command	Code	Note
021	f LABEL 1	201	Label 1: start of each turn
022	RCL n	310	Recall height
023	0	000	
024	f EXP	114	Pause to show height
025	f BRANCH	360	Subroutine 2: show velocity, fuel
026	2	002	
027	START STOP	033	Stop: enter burn
028	JUMP + − =	357	Burn entered? Go to label 3
029	3	003	
030	RCL n	310	No entry: repeat last burn
031	3	003	
032	f LABEL 3	203	Label 3: validate burn
033	f 2	102	Absolute value: no negative burns
034	ST n	300	Store burn in R3
035	3	003	
036	RCL n	310	Recall fuel
037	2	002	
038	RCL n −	312	Subtract burn
039	3	003	
040	JUMP + =	355	Enough fuel? Go to label 4

Register	Purpose	Register	Purpose
0	Lander height	6	
1	Velocity (down = +)	7	
2	Fuel remaining	8	Safe landing speed
3	Burn (also last burn)	9	Lunar gravity
4	Net acceleration A = G - B	.	
5	Scratch (velocity squared)	CHG SIGN	

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Step	Command	Code	Note
041	4	004	
042	RCL n	310	Not enough: burn what is left
043	2	002	
044	ST n	300	Burn = fuel remaining
045	3	003	
046	f LABEL 4	204	Label 4: physics
047	RCL n	310	Recall burn
048	3	003	
049	ST n −	302	Fuel = fuel - burn
050	2	002	
051	RCL n	310	Recall gravity
052	9	011	
053	RCL n −	312	A = gravity - burn
054	3	003	
055	ST n	300	Store A in R4
056	4	004	
057	ST n +	301	Velocity = velocity + A
058	1	001	
059	÷	024	
060	2	002	

Register	Purpose	Register	Purpose
0	Lander height	6	
1	Velocity (down = +)	7	
2	Fuel remaining	8	Safe landing speed
3	Burn (also last burn)	9	Lunar gravity
4	Net acceleration A = G - B	.	
5	Scratch (velocity squared)	CHG SIGN	

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Step	Command	Code	Note
061	=	020	E = A/2
062	RCL$_n$ −	312	E = A/2 - new velocity
063	1	001	
064	ST$_n$ +	301	Height = height - average velocity
065	0	000	
066	RCL$_n$	310	Recall height
067	0	000	
068	JUMP − =	356	Height <= 0? Go to label 5
069	5	005	
070	JUMP	350	Otherwise next turn
071	1	001	
072	f LABEL 5	205	Label 5: touchdown
073	RCL$_n$	310	Recall velocity
074	1	001	
075	f x^2	162	Square it
076	ST$_n$	300	Store in R5
077	5	005	
078	RCL$_n$	310	Recall height (<= 0)
079	0	000	
080	RCL$_n$ ×	313	Multiply by A

Register	Purpose	Register	Purpose
0	Lander height	6	
1	Velocity (down = +)	7	
2	Fuel remaining	8	Safe landing speed
3	Burn (also last burn)	9	Lunar gravity
4	Net acceleration A = G - B	.	
5	Scratch (velocity squared)	CHG SIGN	

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Step	Command	Code	Note
081	4	004	
082	×	023	
083	2	002	
084	=	020	E = 2 x A x height
085	RCL n +	311	Add velocity squared
086	5	005	
087	f 2	102	Absolute value (rounding guard)
088	√	062	Impact speed
089	ST n	300	Store impact speed in R1
090	1	001	
091	RCL n −	312	Subtract safe landing speed
092	8	010	
093	JUMP − =	356	Safe? Go to label 6
094	6	006	
095	5	005	Crash: shows "505." (SOS)
096	0	000	
097	5	005	
098	f 9	111	Identifier
099	f EXP	114	Pause
100	JUMP	350	Go to label 7

Register	Purpose	Register	Purpose
0	Lander height	6	
1	Velocity (down = +)	7	
2	Fuel remaining	8	Safe landing speed
3	Burn (also last burn)	9	Lunar gravity
4	Net acceleration A = G - B	.	
5	Scratch (velocity squared)	CHG SIGN	

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Step	Command	Code	Note
101	7	007	
102	f LABEL 6	206	Label 6: safe landing
103	8	010	Shows "8."
104	f 9	111	Identifier
105	f EXP	114	Pause
106	f LABEL 7	207	Label 7: game over
107	f BRANCH	360	Show impact speed, then fuel (score)
108	2	002	
109	START STOP	033	Stop: game over
110	JUMP	350	Next START begins a new game
111	START STOP	033	(JUMP START/STOP = go to step 001)
112	f LABEL 2	202	Label 2: subroutine
113	RCL n	310	Recall velocity
114	1	001	
115	f EXP	114	Pause to show velocity
116	RCL n	310	Recall fuel (left in display)
117	2	002	
118	f RETURN	030	Return from subroutine
119			
120			

Register	Purpose	Register	Purpose
0	Lander height	6	
1	Velocity (down = +)	7	
2	Fuel remaining	8	Safe landing speed
3	Burn (also last burn)	9	Lunar gravity
4	Net acceleration A = G - B	.	
5	Scratch (velocity squared)	CHG SIGN	

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Step	Command	Code	Note
121			
122			
123			
124			
125			
126			
127			
128			
129			
130			
131			
132			
133			
134			
135			
136			
137			
138			
139			
140			

Register	Purpose	Register	Purpose
0	Lander height	6	
1	Velocity (down = +)	7	
2	Fuel remaining	8	Safe landing speed
3	Burn (also last burn)	9	Lunar gravity
4	Net acceleration A = G - B	.	
5	Scratch (velocity squared)	CHG SIGN	

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Step	Command	Code	Note
141			
142			
143			
144			
145			
146			
147			
148			
149			
150			
151			
152			
153			
154			
155			
156			
157			
158			
159			
160			

Register	Purpose	Register	Purpose
0	Lander height	6	
1	Velocity (down = +)	7	
2	Fuel remaining	8	Safe landing speed
3	Burn (also last burn)	9	Lunar gravity
4	Net acceleration A = G - B	.	
5	Scratch (velocity squared)	CHG SIGN	

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Purpose:
Implements the classic Lunar Lander game, rewritten for the 326 using labels, conditional jumps and a subroutine.
$V_{new} = V_{old}$ + Lunar G - Burn
$H_{new} = H_{old} - V_{avg}$
Try to land (h=0) with an impact speed of 5 or less.
Burns are made positive and limited to the fuel remaining. The landing is judged on the speed at height 0, not at the end of the turn.

Labels:
Step 001 = new game (set up registers)
Label 1 = main loop, one turn per pass
Label 2 = subroutine: show velocity, leave fuel in display
Label 3 = validate burn
Label 4 = physics
Label 5 = touchdown, calculate impact speed
Label 6 = safe landing
Label 7 = game over

Entry conditions:

None. Step 001 sets R0 = 500, R1 = 50, R2 = 120, R8 = 5, R9 = 5.

Custom game: store your own values in R0, R1, R2, R8 and R9, then press JUMP 1 START STOP to skip the set-up.

Execution:

[1] Set the RUN/STEP/LOAD switch to RUN, press JUMP START STOP then START STOP

[2] Height is shown (pause), then velocity (pause)

[3] Program stops with fuel remaining displayed

[4] Enter fuel burn, press START STOP (no entry repeats the last burn)

[5] Repeat from [2] until touchdown

[6] "8." = safe landing, "505." = crash (SOS)

[7] Impact speed is shown (pause), then fuel remaining (score)

[8] Press START STOP for a new game

Exit conditions:
R0 = final height (0 or below)
R1 = impact speed
R2 = fuel remaining (score)

Test game:

Press START STOP with no entry for 6 turns, then burn 73, then 2, then 7. After that press START STOP only.

Heights shown: 500, 447.5, 390, 327.5, 260, 187.5, 110, 64, 50.5, 36.5, 24.5, 14.5, 6.5, 0.5

Touchdown on turn 14: "8.", impact speed 4.7958, fuel 3.